

Production Backtest and Validation Report

Event-Option Static-Union Package

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Abstract

This report summarizes the current production validation evidence for the JEPA event-option static-union package. The active package is `jepa_production_event_options_frozen2025_static_union_202607`, policy `event_option_frozen2025_static_union_balanced_202607`. Results below are the completed-month January–June 2026 production validation used for the July 2026 deployment decision.

1 Production Candidate

The validated package is:

```
neural/models/jepa/jepa_production_event_options_frozen2025_static_union_202607/
```

Result directory:

```
research_papers/JEPA/results/  
event_option_mh30trail_frozen2025_static_union_deploy202607_production_v1
```

The active runtime uses event-option candidate scoring, static policy caps, and mechanical exits. It does not use PPO/RL or OptionValue as the live decision layer.

2 Validation Command

```
python3 neural/jepa/validate_event_option_production_package.py \  
--policy neural/models/jepa/  
    jepa_production_event_options_frozen2025_static_union_202607/event_option_policy  
    .json \  
--registry neural/models/jepa/  
    jepa_production_event_options_frozen2025_static_union_202607/component_registry.  
    json \  
--require-live-ready \  
--ignore-raw-thetadata-coverage \  
--min-profit-factor 1.3 \  
--min-win-rate 0.45 \  
--min-month-trades 12
```

3 Aggregate Metrics

Scope	Trades	Win rate	Profit factor	Min month trades
Overall	697	57.819%	1.915	103
QQQ	236	60.593%	1.771	36
SPXW	355	55.211%	2.037	51
SPY	106	60.377%	1.769	12

Overall validation exceeds the current gates of 45% win rate, 1.3 profit factor, and 12 completed-month trades per ticker. The main limitation is SPY volume. SPY passes the deployable gate but not the preferred 18 trades per month target.

4 Why This Replaced Prior Packages

The previous backfill19 SPY package and level-stability/structural-profile stack were useful stepping stones, but the current static-union package is the deployed artifact because it passes live-ready validation, starts under the current systemd units, and has a clear runtime exit contract:

`stop=-60%/tp=1000%/trail=50%/25%/min_hold=30m/max_hold=180m`

The forced d50 behavior observed in earlier live tests is not the current policy target. The active service units point to the frozen2025 static-union package.

5 Production Caveats

- Validation is only meaningful if the live feed generates the same feature surface used by the package.
- The validation gate is 12 trades per month because SPY does not satisfy 18 trades per month.
- Daily live fills and paper intents are separate evidence and must be audited after deployment.
- A strong backtest does not authorize features that depend on future prices, future option highs/lows, realized PnL, or labels.

6 Live Comparison Protocol

For each live day:

1. Export `rt_data/YYYYMMDD/` and `trades_jepa/`.
2. Parse `paper_order_intents_jepa.jsonl`.
3. Compare selected candidates, entry times, deltas, contracts, exits, and hold times against a replay using the same policy.
4. Run `audit_live_feature_drift.py` after the feed writes `event_option_snapshots_latest.parquet`.

7 Promotion Rule

A future package should replace this one only if it improves the relevant live-ready evidence:

- profit factor and win rate above gate;
- completed-month volume by ticker;
- causal selection with no same-month or future-month leakage;
- live feature equivalence;
- successful systemd startup and smoke test;
- forward live/paper behavior that matches replay.